Customer Behavior Analysis

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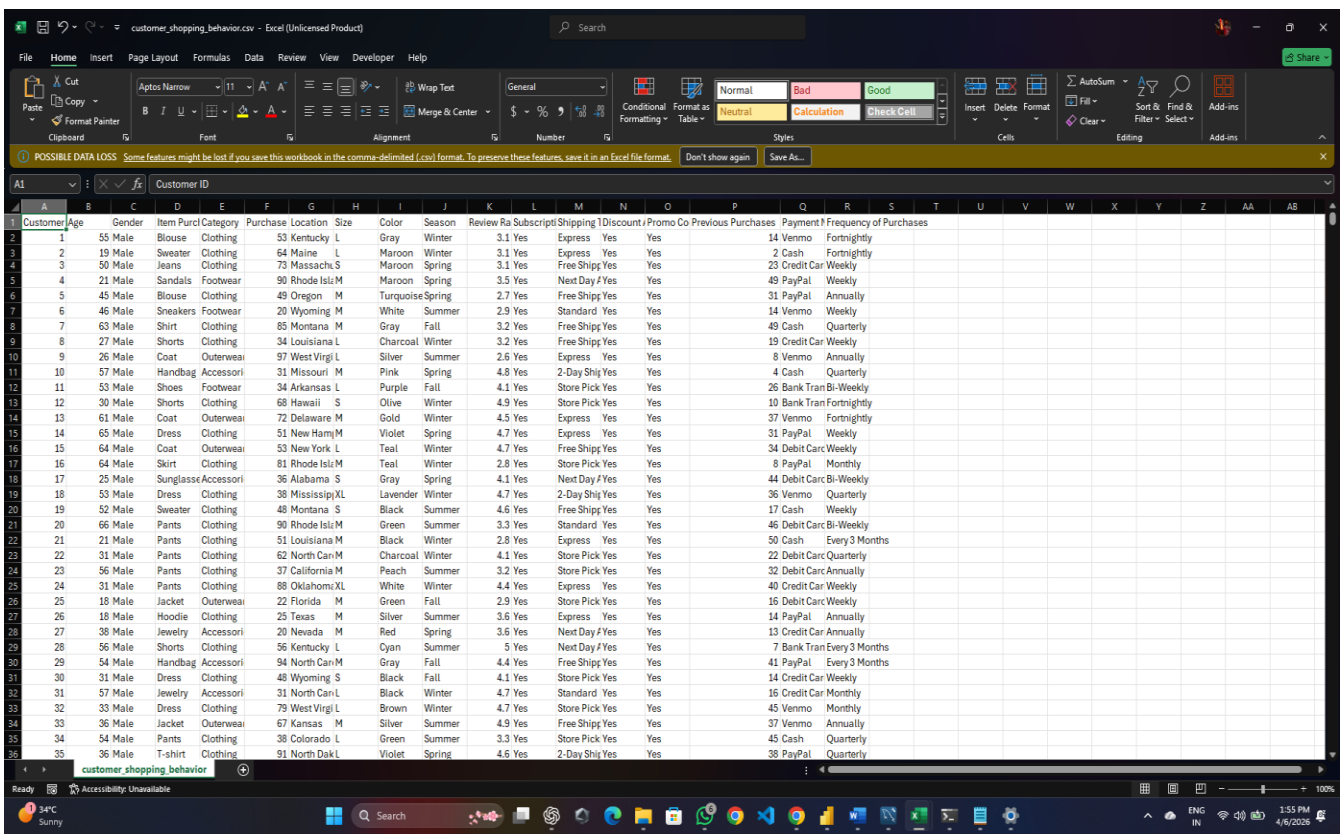
Tools Used: Python (Pandas), SQL, Power BI

Executive Summary

This project focuses on analyzing customer behavior to identify patterns and generate actionable business insights. The dataset was cleaned using Python (Pandas), analyzed using SQL, and visualized using Power BI. The findings highlight customer trends, spending behavior, and product performance, helping improve decision-making and business strategy.

Dataset Description

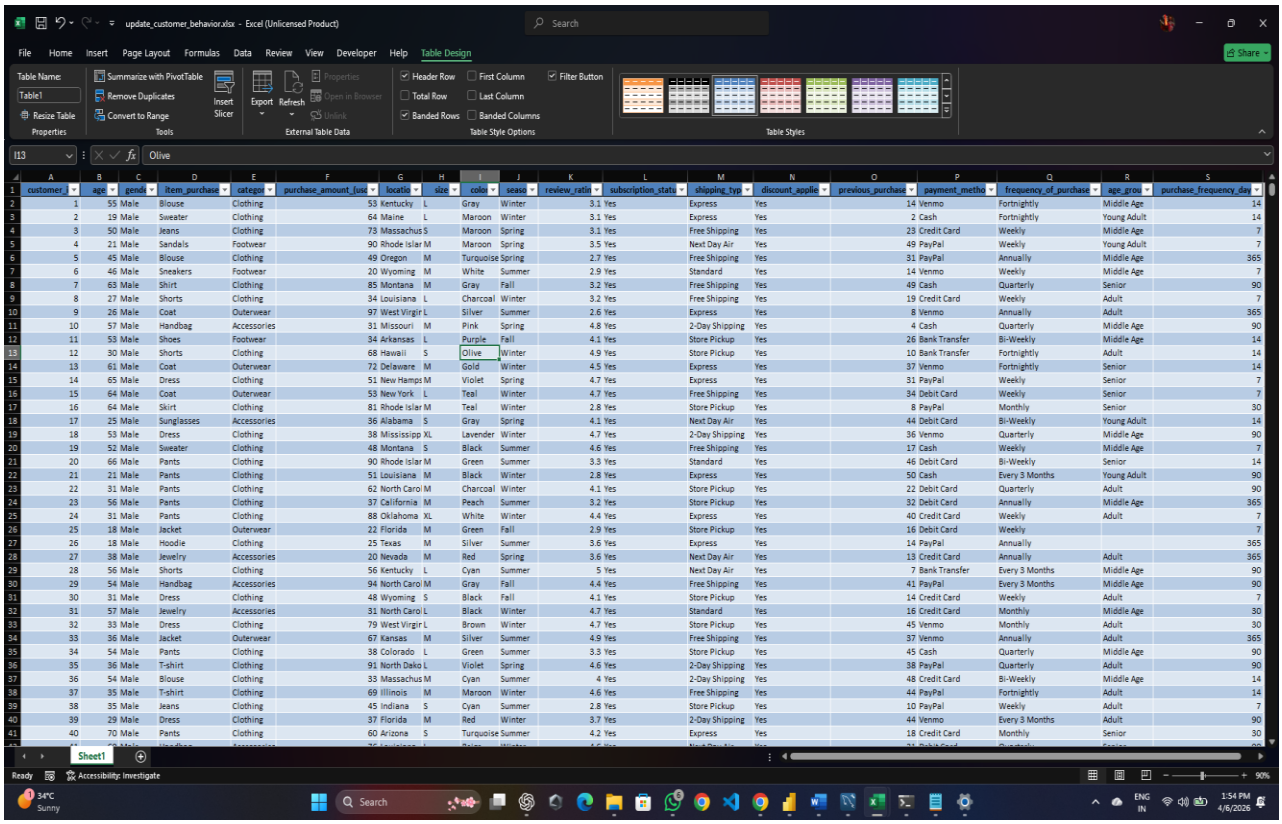
The dataset contains customer transaction data including attributes such as Customer ID, Gender, Age Group, Item Purchased, Category, Purchase Amount, Review Rating, Shipping Type, Discount Applied, Subscription Status, and Previous Purchases.



Customer	Age	Gender	Item Purcl	Category	Purchase	Location	Size	Color	Season	Review Ra	Subscripti	Shipping T	Discount /	Promo Co	Previous Purchases	Payment M	Frequency of Purchases
1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
3	50	Male	Jeans	Clothing	73	Massachu	S	Maroon	Spring	3.1	Yes	Free Shipp	Yes	Yes	23	Credit Car	Weekly
4	21	Male	Sandals	Footwear	90	Rhode Isla	M	Maroon	Spring	3.5	Yes	Next Day A	Yes	Yes	49	PayPal	Weekly
5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipp	Yes	Yes	31	PayPal	Annually
6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes	Yes	14	Venmo	Weekly
7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipp	Yes	Yes	49	Cash	Quarterly
8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipp	Yes	Yes	19	Credit Car	Weekly
9	26	Male	Coat	Outerwear	97	West Virgi	L	Silver	Summer	2.6	Yes	Express	Yes	Yes	8	Venmo	Annually
10	57	Male	Handbag	Accessori	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Ship	Yes	Yes	4	Cash	Quarterly
11	53	Male	Shoes	Footwear	34	Arkansas	L	Purple	Fall	4.1	Yes	Store Pick	Yes	Yes	26	Bank Tran	Bi-Weekly
12	30	Male	Shorts	Clothing	68	Hawaii	S	Olive	Winter	4.9	Yes	Store Pick	Yes	Yes	10	Bank Tran	Fortnightly
13	61	Male	Coat	Outerwear	72	Delaware	M	Gold	Winter	4.5	Yes	Express	Yes	Yes	37	Venmo	Fortnightly
14	65	Male	Dress	Clothing	51	New Hamp	M	Violet	Spring	4.7	Yes	Express	Yes	Yes	31	PayPal	Weekly
15	64	Male	Coat	Outerwear	53	New York	L	Teal	Winter	4.7	Yes	Free Shipp	Yes	Yes	34	Debit Carc	Weekly
16	64	Male	Skirt	Clothing	81	Rhode Isla	M	Teal	Winter	2.8	Yes	Store Pick	Yes	Yes	8	PayPal	Monthly
17	25	Male	Sunglasse	Accessori	36	Alabama	S	Gray	Spring	4.1	Yes	Next Day A	Yes	Yes	44	Debit Carc	Bi-Weekly
18	53	Male	Dress	Clothing	38	Mississipp	XL	Lavender	Winter	4.7	Yes	2-Day Ship	Yes	Yes	36	Venmo	Quarterly
19	52	Male	Sweater	Clothing	48	Montana	S	Black	Summer	4.6	Yes	Free Shipp	Yes	Yes	17	Cash	Weekly
20	66	Male	Pants	Clothing	90	Rhode Isla	M	Green	Summer	3.3	Yes	Standard	Yes	Yes	46	Debit Carc	Bi-Weekly
21	21	Male	Pants	Clothing	51	Louisiana	M	Black	Winter	2.8	Yes	Express	Yes	Yes	50	Cash	Every 3 Months
22	31	Male	Pants	Clothing	62	North Car	M	Charcoal	Winter	4.1	Yes	Store Pick	Yes	Yes	22	Debit Carc	Quarterly
23	56	Male	Pants	Clothing	37	California	M	Peach	Summer	3.2	Yes	Store Pick	Yes	Yes	32	Debit Carc	Annually
24	31	Male	Pants	Clothing	88	Oklahoma	XL	White	Winter	4.4	Yes	Express	Yes	Yes	40	Credit Car	Weekly
25	18	Male	Jacket	Outerwear	22	Florida	M	Green	Fall	2.9	Yes	Store Pick	Yes	Yes	16	Debit Carc	Weekly
26	18	Male	Hoodie	Clothing	25	Texas	M	Silver	Summer	3.6	Yes	Express	Yes	Yes	14	PayPal	Annually
27	38	Male	Jewelry	Accessori	20	Nevada	M	Red	Spring	3.6	Yes	Next Day A	Yes	Yes	13	Credit Car	Annually
28	56	Male	Shorts	Clothing	56	Kentucky	L	Cyan	Summer	5	Yes	Next Day A	Yes	Yes	7	Bank Tran	Every 3 Months
29	54	Male	Handbag	Accessori	94	North Car	M	Gray	Fall	4.4	Yes	Free Shipp	Yes	Yes	41	PayPal	Every 3 Months
30	31	Male	Dress	Clothing	48	Wyoming	S	Black	Fall	4.1	Yes	Store Pick	Yes	Yes	14	Credit Car	Weekly
31	57	Male	Jewelry	Accessori	31	North Car	L	Black	Winter	4.7	Yes	Standard	Yes	Yes	16	Credit Car	Monthly
32	33	Male	Dress	Clothing	79	West Virgi	L	Brown	Winter	4.7	Yes	Store Pick	Yes	Yes	45	Venmo	Monthly
33	36	Male	Jacket	Outerwear	67	Kansas	M	Silver	Summer	4.9	Yes	Free Shipp	Yes	Yes	37	Venmo	Annually
34	54	Male	Pants	Clothing	38	Colorado	L	Green	Summer	3.3	Yes	Store Pick	Yes	Yes	45	Cash	Quarterly
35	36	Male	T-shirt	Clothing	91	North Dak	L	Violet	Spring	4.6	Yes	2-Day Ship	Yes	Yes	38	PayPal	Quarterly

Data Cleaning (Pandas)

Data cleaning was performed using Pandas. Missing values were handled, duplicates were removed, data types were corrected, and new columns such as age groups were created.



customer_	age	gend	item_purchase	categor	purchase_amount_(us	locatio	size	color	seaso	review_ratin	subscription_stats	shipping_typ	discount_applie	previous_purchase	payment_metho	frequency_of_purchase	age_gro	purchase_frequency_day
1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	14	Venmo	Fortnightly	Middle Age	14
2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	2	Cash	Fortnightly	Young Adult	14
3	50	Male	Jeans	Clothing	73	Massachus	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	23	Credit Card	Weekly	Middle Age	7
4	21	Male	Sandals	Footwear	90	Rhode Islar	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	49	PayPal	Weekly	Young Adult	7
5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	31	PayPal	Annually	Middle Age	365
6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes	14	Venmo	Weekly	Middle Age	7
7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping	Yes	49	Cash	Quarterly	Senior	90
8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping	Yes	19	Credit Card	Weekly	Adult	7
9	26	Male	Coat	Outerwear	97	West Virgir	L	Silver	Summer	2.6	Yes	Express	Yes	8	Venmo	Annually	Adult	365
10	57	Male	Handbag	Accessories	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping	Yes	4	Cash	Quarterly	Middle Age	90
11	53	Male	Shoes	Footwear	34	Arkansas	L	Purple	Fall	4.1	Yes	Store Pickup	Yes	26	Bank Transfer	Bi-Weekly	Middle Age	14
12	30	Male	Shorts	Clothing	68	Hawaii	S	Olive	Winter	4.9	Yes	Store Pickup	Yes	10	Bank Transfer	Fortnightly	Adult	14
13	61	Male	Coat	Outerwear	72	Delaware	M	Gold	Winter	4.5	Yes	Express	Yes	37	Venmo	Fortnightly	Senior	14
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17	25	Male	Sunglasses	Accessories	36	Alabama	S	Gray	Spring	4.1	Yes	Next Day Air	Yes	44	Debit Card	Bi-Weekly	Young Adult	14
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28	56	Male	Shorts	Clothing	56	Kentucky	L	Cyan	Summer	5	Yes	Next Day Air	Yes	7	Bank Transfer	Every 3 Months	Middle Age	90
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36	54	Male	Blouse	Clothing	33	Massachus	M	Cyan	Summer	4	Yes	2-Day Shipping	Yes	48	Credit Card	Bi-Weekly	Middle Age	14
37	35	Male	T-shirt	Clothing	69	Illinois	M	Maroon	Winter	4.6	Yes	Free Shipping	Yes	44	PayPal	Fortnightly	Adult	14
38	35	Male	Jeans	Clothing	45	Indiana	S	Cyan	Summer	2.8	Yes	Store Pickup	Yes	10	PayPal	Weekly	Adult	7
39	29	Male	Dress	Clothing	37	Florida	M	Red	Winter	3.7	Yes	2-Day Shipping	Yes	44	Venmo	Every 3 Months	Adult	90
40	70	Male	Pants	Clothing	60	Arizona	S	Turquoise	Summer	4.2	Yes	Express	Yes	18	Credit Card	Monthly	Senior	30

Problem Statement

The project addresses the following business questions:

- What is the total revenue generated by male vs. female customers?
- select sum(`purchase_amount_(usd)`), gender from customer group by(gender);Identify customers who used discounts but still spent above average

Result Grid | Filter Rows:

sum(`purchase_amount_(usd)`)	gender
157890	Male
75191	Female

- Which customers used a discount but still spent more than the average purchase amount?

- select customer_id , `purchase_amount_(usd)` from customer where discount_applied = 'Yes' and `purchase_amount_(usd)` > (select avg(`purchase_amount_(usd)`) from customer);

customer_id	purchase_amount_(usd)
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62

- Which are the top 5 products with the highest average review rating?

- select item_purchased, round(avg(review_rating),2) as avg_ratings from customer group by item_purchased order by avg_ratings desc limit 10;Evaluate whether subscription impacts customer spending

item_purchased	avg_ratings
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78
Skirt	3.78
T-shirt	3.78
Sweater	3.76
Sneakers	3.76
Jacket	3.76

- Compare the average Purchase Amounts between Standard and Express Shipping.

select shipping_type , round(avg(`purchase_amount_(usd)`),2) as avg_purchase_amount from customer where shipping_type in ('Standard', 'Express') group by shipping_type;

item_purchased	avg_ratings
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78
Skirt	3.78
T-shirt	3.78
Sweater	3.76
Sneakers	3.76
Jacket	3.76

- Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

```
Select subscription_status, round(avg(`purchase_amount_(usd)`),2) as avg_total_spend,
count(customer_id) as total_customer,  round(sum(`purchase_amount_(usd)`),2) as
total_revenue from customer group by subscription_status;
```

subscription_status	avg_total_spend	total_customer	total_revenue
Yes	59.49	1053	62645
No	59.87	2847	170436

- Which 5 products have the highest percentage of purchases with discounts applied?

```
select item_purchased as products, Round((100 * sum(case when discount_applied =
'Yes' Then 1 Else 0 END))/count(*),2) as total_buying  from customer group by
item_purchased order by total_buying desc limit 5;
```

products	total_buying
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

- Segment customers into New, Returning, and Loyal based on their total number of previous

- purchases, and show the count of each segment.

```
with customer_type as (
select customer_id, previous_purchases,
case
when previous_purchases = 1 then 'New'
when previous_purchases between 2 and 10 then 'Returning'
when previous_purchases >= 10 then 'Loyal'
end as customer_segment
from customer
)
select customer_segment, count(*) as 'Number of Customer'
from customer_type
group by customer_segment;
```

customer_segment	Number of Customer
Loyal	3116
Returning	701
New	83

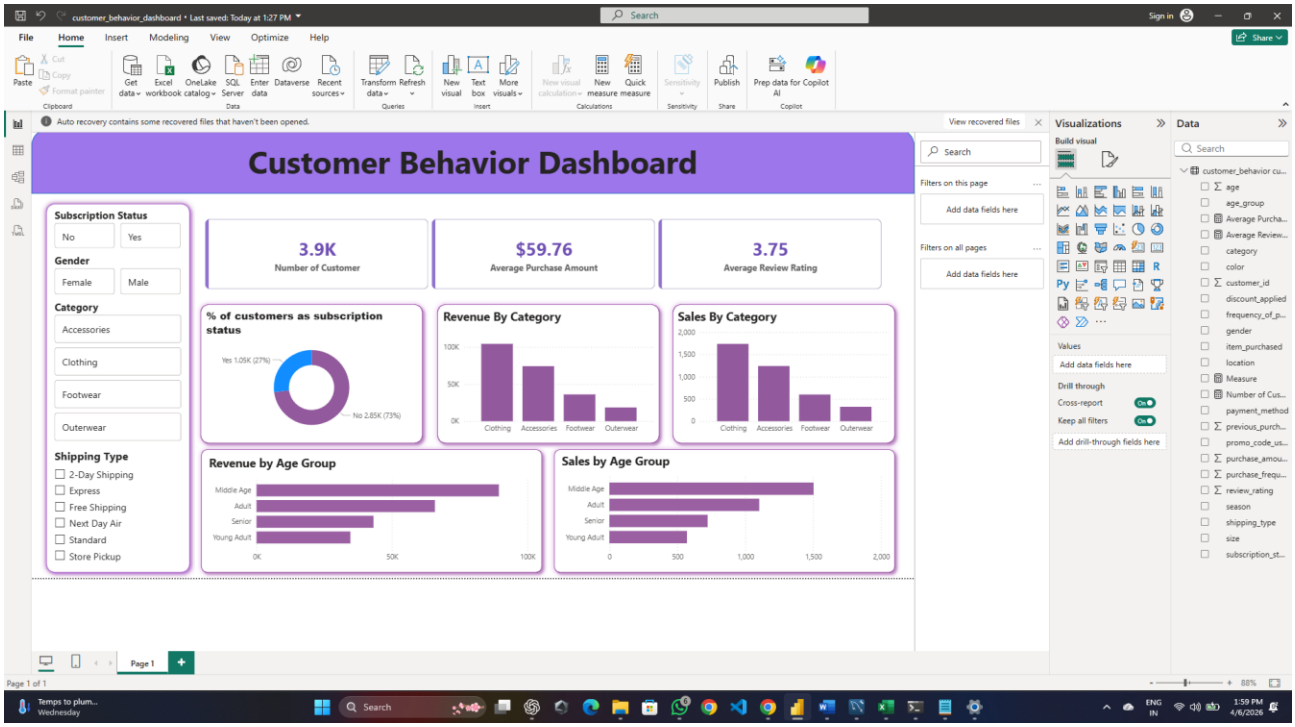
- What are the top 3 most purchased products within each category?

```
- with most_purchased_products as (
select category, item_purchased, count(customer_id) as total_customer,
row_number() over(partition by category order by count(customer_id) desc) as
rank_of_product
from customer group by category, item_purchased
)
select category, item_purchased, total_customer, rank_of_product from
most_purchased_products
where rank_of_product <= 3;
```

category	item_purchased	total_customer	rank_of_product
Accessories	Jewelry	171	1
Accessories	Sunglasses	161	2
Accessories	Belt	161	3
Clothing	Blouse	171	1
Clothing	Pants	171	2
Clothing	Shirt	169	3
Footwear	Sandals	160	1
Footwear	Shoes	150	2
Footwear	Sneakers	145	3
Outerwear	Jacket	163	1
Outerwear	Coat	161	2

Data Visualization (Power BI)

An interactive dashboard was created showing KPIs, revenue trends, product performance, customer segmentation, and subscription analysis.



Key Insights

1. Revenue Contribution by Gender

The analysis shows a difference in total revenue contribution between male and female customers. One segment contributes a higher share of total revenue, indicating stronger purchasing behavior.

Business Impact: Marketing strategies can be tailored toward the higher-spending gender while identifying opportunities to improve engagement in the other segment.

2. High-Value Customers Using Discounts

A group of customers used discounts but still made purchases above the average spending level.

Business Impact: Discounts are effective not only for attracting low spenders but also for encouraging higher spending among valuable customers. Strategic discounting can boost overall revenue.

3. Top Performing Products (Based on Ratings)

Certain products consistently achieved the highest average review ratings.

Business Impact: These products indicate high customer satisfaction and should be prioritized in marketing campaigns and inventory planning.

4. Impact of Shipping Type on Spending

A noticeable difference exists in average purchase amounts between Standard and Express shipping users.

Business Impact: Customers opting for faster delivery tend to have different spending behavior, suggesting opportunities to promote premium shipping options.

5. Subscription Increases Customer Value

Subscribed customers show higher average spending and contribute more to total revenue compared to non-subscribers.

Business Impact: Subscription models improve customer retention and lifetime value. Encouraging subscriptions can significantly enhance revenue.

6. Discount Sensitivity by Product

Some products have a higher percentage of purchases where discounts were applied.

Business Impact: These products are more price-sensitive, and discounts play a key role in driving their sales.

7. Customer Segmentation Insights

Customers were categorized into New, Returning, and Loyal segments. Loyal customers, although fewer in number, contribute significantly to total revenue.

Business Impact: Retention strategies should focus on converting customers into loyal segments to maximize long-term profitability.

8. Category-Wise Product Dominance

Within each category, a few products dominate in terms of purchase frequency.

Business Impact: Businesses should focus on these top-performing products for promotions, inventory optimization, and strategic placement.

9. Repeat Buyers and Subscription Behavior

Customers with higher previous purchases show a tendency to be subscribed.

Business Impact: There is a strong relationship between customer loyalty and subscription adoption, which can be leveraged for targeted campaigns.

10. Revenue Contribution by Age Group

Certain age groups contribute significantly more to total revenue than others.

Business Impact: Marketing strategies should be tailored toward high-performing age groups to maximize revenue.

Conclusion

This project successfully analyzed customer behavior using a structured data analysis approach involving data cleaning, querying, and visualization. The dataset was first preprocessed using Python (Pandas) to ensure accuracy and consistency, followed by SQL-based analysis to answer key business questions. Finally, Power BI was used to create an interactive dashboard for clear and effective visualization of insights.

The analysis revealed important patterns in customer purchasing behavior, including the impact of discounts, subscription status, product preferences, and demographic factors such as age and gender. It was observed that certain customer segments contribute significantly more to revenue, while loyal and subscribed customers play a crucial role in overall business performance.

Overall, this project demonstrates how raw data can be transformed into meaningful insights that support data-driven decision-making. The findings can help businesses optimize their marketing strategies, improve customer retention, and enhance revenue generation.

Project Files:

- - Dataset (CSV file)

customer_shopping
_behavior.csv

- - SQL Queries

**customer_behavior.
sql**

- Power BI Dashboard

**customer_behavior_
dashboard.pbix**